

Predix – A Diabetes Risk Scoring System

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Overview

Predix is an Android application, backed by a Flask REST API and a trained Random Forest model, that estimates a user's diabetes risk from everyday health indicators and explains the result with feature-level breakdowns. Unlike most existing risk calculators, which return a single opaque score, Predix shows exactly which health indicators are driving the prediction, alongside a running history of past assessments and evidence-based health tips.

Key Features

- Real-time risk prediction — a trained Random Forest model (80 trees, 21 features) returns a classification and probability score within seconds.
- Explainable results — each prediction is broken down into top contributing factors (e.g. BMI, blood pressure, general health) with directional insight.
- Lab report analysis — users can enter lab values (HbA1c, fasting/post-meal glucose, LDL) directly or upload a report image for AI-assisted interpretation.
- Patient history — every prediction is logged with patient ID, BMI, age bracket, and risk percentage for longitudinal tracking.
- Health indicators checklist and curated, cited lifestyle guidance sourced from bodies such as the CDC, ADA, and Harvard Health.

Technologies Used

Android Studio (WebView-based UI), Chart.js · Python, Flask, Flask-CORS, Gunicorn · scikit-learn (Random Forest), NumPy, joblib · Anthropic Claude API (lab report image analysis) · Render.com (deployment) · Git & GitHub.

Machine Learning Model

A Random Forest Classifier (scikit-learn) is trained on the Kaggle “Diabetes Health Indicators Dataset (BRFSS 2015),” comprising roughly 229K patient records across 21 demographic, lifestyle, and clinical features. Continuous features are standardized with a fitted StandardScaler, and BMI is clipped at 46.0 to limit outlier influence. The model outputs one of three classes — No Diabetes, Pre-Diabetic, or Diabetic — and achieved a reported test

ROC-AUC of 0.73, selected as the best-performing model over Logistic Regression and XGBoost on accuracy, recall, and F1-score.

API Reference

The Flask backend (app.py) exposes four endpoints: **GET** / for a health check; **POST** /**predict**, which accepts the 21 health features as JSON and returns the risk classification, probability, and top contributing factors; **POST** /**analyze**, which accepts lab values or a base64-encoded report image and returns a plain-language explanation; and **POST** /**iot/update** / **GET** /**iot/latest**, which push and fetch BMI or blood-pressure readings from a connected device.

Project Structure

predix/ — app.py (Flask backend), requirements.txt, render.yaml — model/ (diabetes_model.pkl, scaler.pkl, model_metadata.json) — android/ (Predix.html, WebView UI) — docs/FinalReport_8th_sem.pdf.

Getting Started

Clone the repository and install dependencies with **pip install -r requirements.txt**. Optionally set **ANTHROPIC_API_KEY** for AI-assisted lab report analysis, then run **python app.py** to serve the API at <http://localhost:5000>. Deployment on Render.com runs gunicorn app:app with two workers via the included render.yaml, with the API key set manually in the Render dashboard. For the mobile app, open android/ in Android Studio, point the base URL to the running Flask instance, and build for Android 8.0 or later.

Results & Future Work

End-to-end usability testing showed low-latency responses between form submission and result display. Planned work includes a fully native Android experience, wearable/IoT integration, a personalized diet and exercise recommendation engine, expansion to a multi-disease risk engine (hypertension, cardiovascular disease, stroke), and multi-class diabetes subtype classification.

Disclaimer

Predix is an educational and decision-support tool developed as part of an academic major project. It is **not** a certified medical device, and its predictions should not substitute for professional medical diagnosis. Always consult a qualified healthcare provider regarding any health concerns.